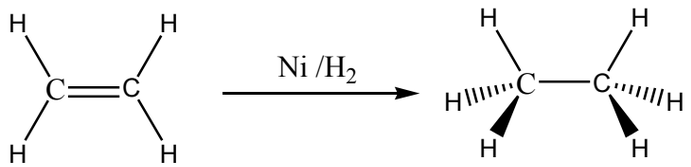
Ion	Configuration	n	$\mu = \sqrt{n(n+2)}\ \mu_B$	$\mu_{(observed)}$
Sc^{3+} ,Ti^{4+} ,V^{5+}	d^0	0	$\mu = \sqrt{0(0+2)} = 0\ \mu_B$	diamagnetic
Ti^{3+}, V^{4+}	d^1	1	$\mu = \sqrt{1(1+2)} = \sqrt{3} = 1.73\ \mu_B$	1.75
Ti^{2+}, V^{3+}	d^2	2	$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.83\ \mu_B$	2.76
Cr^{3+}, Mn^{4+}, V^{2+}	d^3	3	$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87\ \mu_B$	3.86
Cr^{2+}, Mn^{3+}	d^4	4	$\mu = \sqrt{4(4+2)} = \sqrt{24} = 4.89\ \mu_B$	4.80
Mn^{2+}, Fe^{3+}	d^5	5	$\mu = \sqrt{5(5+2)} = \sqrt{35} = 5.91\ \mu_B$	5.96
Co^{3+}, Fe^{2+}	d^6	4	$\mu = \sqrt{4(4+2)} = \sqrt{24} = 4.89\ \mu_B$	5.3-5.5
Co^{2+}	d^7	3	$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87\ \mu_B$	4.4-5.2
Ni^{2+}	d^8	2	$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.83\ \mu_B$	2.9-3.4
Cu^{2+}	d^9	1	$\mu = \sqrt{1(1+2)} = \sqrt{3} = 1.732\ \mu_B$	1.8-2.2
Cu^{+} , Zn^{2+}	d^{10}	0	$\mu = \sqrt{0(0+2)} = 0\ \mu_B$	diamagnetic

4.3.7 Catalytic properties

The chemical industries manufacture a number of products such as polymers, flavours, drugs etc., Most of the manufacturing processes have adverse effect on the environment so there is an interest for eco friendly alternatives. In this context, catalyst based manufacturing processes are advantageous, as they require low energy, minimize waste production and enhance the conversion of reactants to products.

Many industrial processes use transition metals or their compounds as catalysts. Transition metal has energetically available d orbitals that can accept electrons from reactant molecule or metal can form bond with reactant molecule using its d electrons. For example, in the catalytic hydrogenation of an alkene, the alkene bonds to an active site by using its π electrons with an empty d orbital of the catalyst.

The σ bond in the hydrogen molecule breaks, and each hydrogen atom forms a bond with a d electron on an atom in the catalyst. The two hydrogen atoms then bond with the partially broken π -bond in the alkene to form an alkane.



Alkene Alkane

In certain catalytic processes the variable oxidation states of transition metals find applications. For example, in the manufacture of sulphuric acid from SO_3, vanadium pentoxide

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